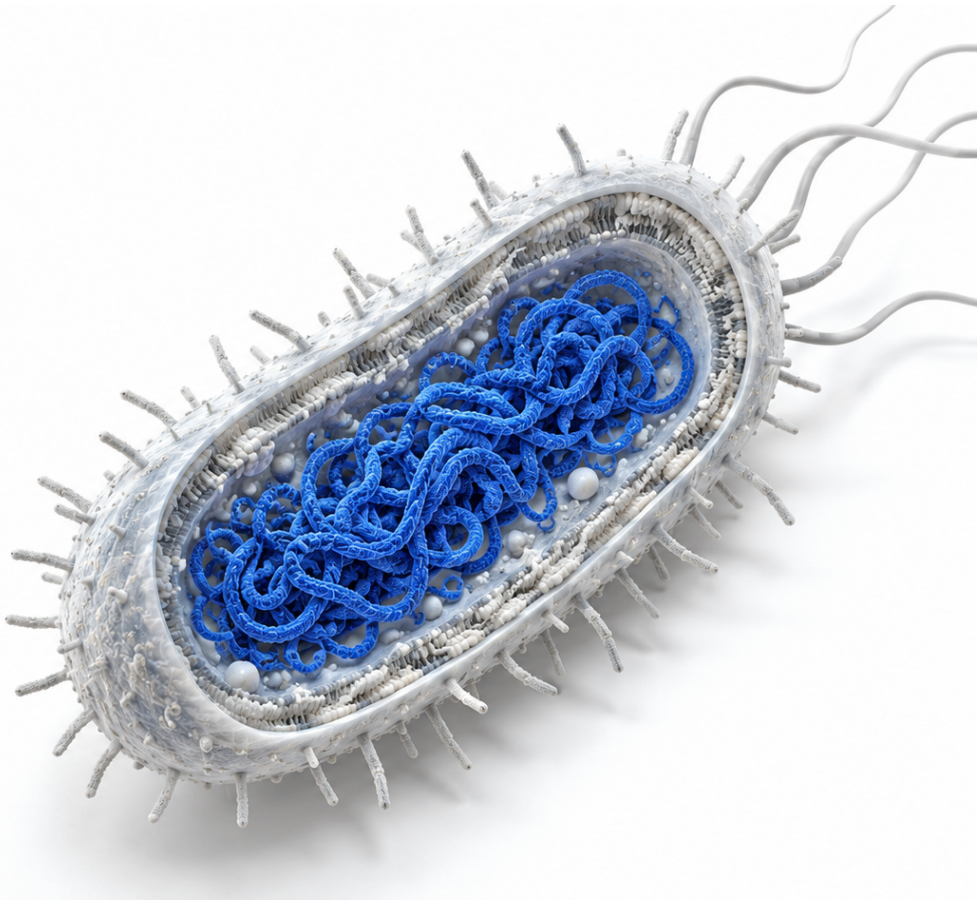
E. coli

Variant analysis

From paired-end reads to comparative evidence



Ziad

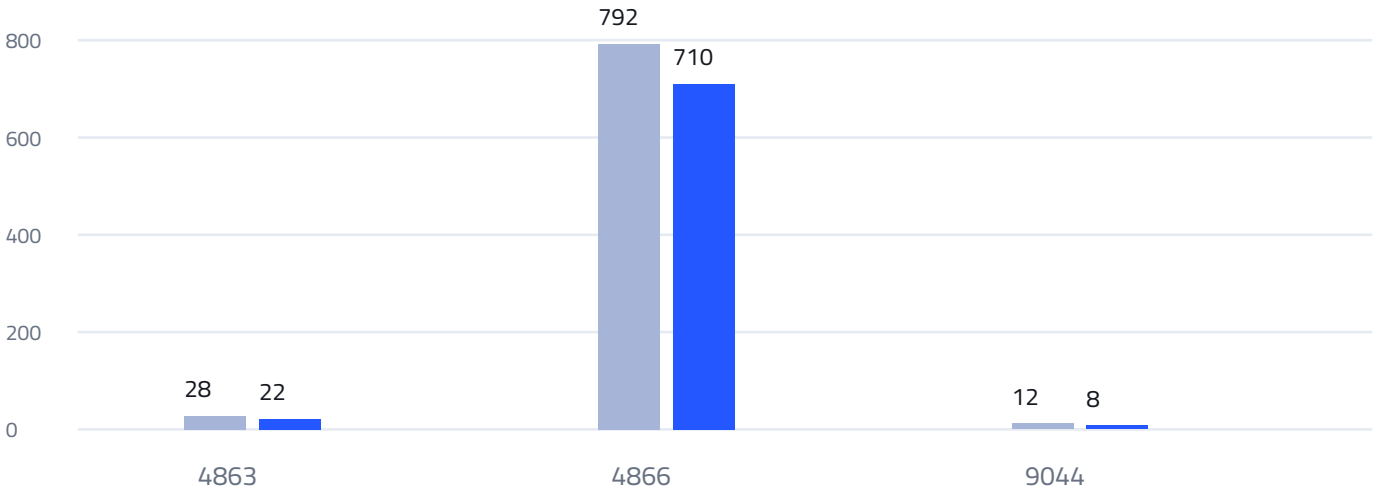
REL606 / CP000819.1 / THREE SAMPLES

Conceptual cutaway illustration; not experimental microscopy.

The results at a glance

Three paired-end E. coli samples were aligned against REL606. A haploid variant-calling model, consistent filtering and exact-allele comparison produced the following retained call sets.

22	710	8
SRR2584863	SRR2584866	SRR2589044



Raw (gray) and filtered (blue) variant records; counts from VCF files.

Six shared variants; one coverage investigation

Five SNPs and one two-base deletion were shared by all three filtered sets. All six overlap coding regions. Manual sequence checks predict five missense substitutions and one frameshift with an early stop.

SRR2584866 had a 43,944-base zero-depth interval. The gap is observed; a deletion junction was not established. Variant counts alone do not establish mutation rate, selection or direct ancestry.

Evidence basis: original FASTQ, reference, BAM, VCF, QC, statistical reports and saved IGV figures. Commands below consolidate the documented procedure; they are not a verbatim terminal transcript.

01 / Inputs and analysis environment

The ecoli Conda environment on Ubuntu provided the tools. Relative paths below start in the project directory. Each sample contains 175,000 read pairs, represented by an R1 and an R2 FASTQ.

```
conda activate ecoli
cd ~/Desktop/ecoli_variant_project
mkdir -p data/reads data/ref_genome results/bam results/vcf
mkdir -p results/qc/raw reports
curl -fL -o data/reads/sub.tar.gz \
  https://ndownloader.figshare.com/files/14418248
tar -xzf data/reads/sub.tar.gz -C data/reads
```

curl downloaded the archive and tar extracted it. The supplied trim.sub reads were already trimmed and subsampled; no additional Trimmomatic step was performed.

Sample	FASTQ files	Read pairs
SRR2584863	R1 + R2	175,000
SRR2584866	R1 + R2	175,000
SRR2589044	R1 + R2	175,000

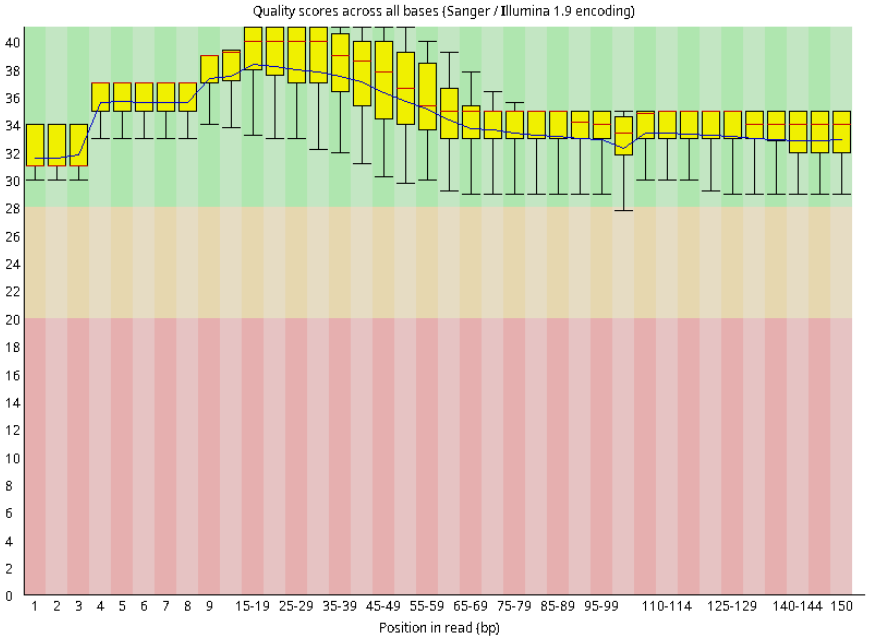
```
data/reads/sub/<sample>_1.trim.sub.fastq
data/reads/sub/<sample>_2.trim.sub.fastq
```

Here <sample> denotes one of the three accessions, not literal filename text. Tool versions: BWA 0.7.19, SAMtools 1.24, BCFtools 1.24, VCFtools 0.1.17, FastQC 0.13.0 and MultiQC 1.35. IGV was used for visual inspection.

02 / Quality assessment

```
fastqc -t 2 --outdir results/qc/raw data/reads/sub/*.fastq
multiqc results/qc/raw --outdir results/qc/raw
```

FastQC assessed each FASTQ; MultiQC aggregated all six reports. Neither tool modified the reads. Individual HTML/ZIP files and the MultiQC report supplied the evidence.



Original FastQC per-base-quality plot: SRR2584863 R2.

All six files passed base quality, read quality, N content, adapter content, overrepresented sequences and duplication-level checks. All received WARN for GC and length distribution.

Per-base composition failed for 4863 R2, 4866 R1 and 9044 R2, and warned for the other files. These flags were documented; they did not alone establish contamination or unusable reads. No second trimming or post-trimming QC was generated.

```
results/qc/raw/*_fastqc.html
results/qc/raw/*_fastqc.zip
results/qc/raw/multiqc_report.html
results/qc/raw/multiqc_data/
```

03 / Reference preparation

4,629,812

REFERENCE BASES

Reference: E. coli B strain REL606, chromosome CP000819.1, assembly GCA_000017985.1_ASM1798v1. The same sequence was used for alignment, variant calling and annotation.

```
base=https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/017/985
assembly=GCA_000017985.1_ASM1798v1
url="$base/$assembly/${assembly}_genomic.fna.gz"
curl -fL -o data/ref_genome/ecoli_rel606.fasta.gz "$url"
gzip -t data/ref_genome/ecoli_rel606.fasta.gz
gunzip -k data/ref_genome/ecoli_rel606.fasta.gz
bwa index data/ref_genome/ecoli_rel606.fasta
samtools faidx data/ref_genome/ecoli_rel606.fasta
```

gzip -t checked compressed integrity; gunzip -k retained the compressed source. bwa index created alignment indexes, while faidx created the FASTA coordinate-access index.

```
data/ref_genome/ecoli_rel606.fasta
data/ref_genome/ecoli_rel606.fasta.fai
# BWA indexes: .amb .ann .bwt .pac .sa
```

Basis: FASTA header, .fai sequence length, BAM headers and coverage reports agree on CP000819.1 and its length. The URL variables above are a readable equivalent of the download command.

04 / Paired-end alignment

BWA-MEM aligned both read ends against REL606 with two threads. Read groups assigned a distinct sample name to each accession and recorded the ILLUMINA platform.

```
sample=SRR2584863
ref=data/ref_genome/ecoli_rel606.fasta
bwa mem -t 2 \
  -R "@RG\tID:${sample}\tSM:${sample}\tPL:ILLUMINA" \
  "$ref" \
  "data/reads/sub/${sample}_1.trim.sub.fastq" \
  "data/reads/sub/${sample}_2.trim.sub.fastq" \
  > "results/bam/${sample}.sam"
samtools view -b -o "results/bam/${sample}.bam" \
  "results/bam/${sample}.sam"
samtools quickcheck -v "results/bam/${sample}.bam"
```

The same procedure was repeated for SRR2584866 and SRR2589044. SAM is the text alignment format; BAM is its binary counterpart. quickcheck performed a preliminary header/end-of-file check, not full record validation.

Sample	Primary reads	Alignment records
SRR2584863	350,000	352,017
SRR2584866	350,000	351,169
SRR2589044	350,000	353,820

Extra records are supplementary alignments, not additional sequenced reads. BAM headers retain the reference and program commands. Outputs: results/bam/<sample>.sam and <sample>.bam.

05 / BAM processing

```
samtools collate -o "results/bam/${sample}.collated.bam" \
  "results/bam/${sample}.bam"
samtools fixmate -m "results/bam/${sample}.collated.bam" \
  "results/bam/${sample}.fixmate.bam"
samtools sort -o "results/bam/${sample}.sorted.bam" \
  "results/bam/${sample}.fixmate.bam"
samtools markdup -s "results/bam/${sample}.sorted.bam" \
  "results/bam/${sample}.markdup.bam"
samtools index "results/bam/${sample}.markdup.bam"
```

Why this order matters

collate groups records by read name. fixmate -m updates mate information and adds the mate-score tag. sort then orders records by genomic coordinate, allowing markdup to flag potential duplicates. index enables regional retrieval.

Sample	Duplicate records	Records removed
SRR2584863	592	0
SRR2584866	547	0
SRR2589044	642	0

markdup was used without -r, so duplicate records remained in BAM. Counts agree with flagstat and direct duplicate-flag counting. Coordinate ordering and indexed region retrieval were checked in the original file review.

```
results/bam/<sample>.collated.bam
results/bam/<sample>.fixmate.bam
results/bam/<sample>.sorted.bam
results/bam/<sample>.markdup.bam
results/bam/<sample>.markdup.bam.bai
```

06 / Alignment and genome coverage

```
samtools flagstat "results/bam/${sample}.markdup.bam" \
  > "reports/${sample}.flagstat.txt"
samtools coverage "results/bam/${sample}.markdup.bam" \
  > "reports/${sample}.coverage.tsv"
```

flagstat summarizes alignment flags; coverage measures covered reference bases, mean depth and quality. The saved reports matched recalculation from the original BAM files.

Sample	Mapped %	Proper pairs %	Duplicates
4863	99.96	98.03	592
4866	99.98	99.05	547
9044	99.94	96.33	642

Sample	Covered bases	Coverage %	Mean depth
4863	4,620,058	99.7893	9.89791
4866	4,306,683	93.0207	9.95870
9044	4,620,907	99.8077	9.96615

Sample	Mean base quality	Mean mapping quality
4863	35.3	58.6
4866	35.4	58.2
9044	35.4	58.6

Mapped % refers to primary reads. Mean depth near 10x does not mean ten reads at every site. SRR2584866 combined the highest mapping percentage with the lowest reference coverage, motivating the gap investigation.

07 / Calling and filtering variants

```
bcftools mpileup -f "$ref" -a FORMAT/DP,FORMAT/AD \
  -Ou "results/bam/${sample}.markdup.bam" | \
  bcftools call --ploidy 1 -mv -Oz \
  -o "results/vcf/${sample}.raw.vcf.gz"
vcftools --gzvcf "results/vcf/${sample}.raw.vcf.gz" \
  --minQ 30 --minDP 5 --max-missing 1 \
  --recode --recode-INFO-all \
  --out "results/vcf/${sample}.filtered"
bcftools view -Oz -o "results/vcf/${sample}.filtered.vcf.gz" \
  "results/vcf/${sample}.filtered.recode.vcf"
bcftools index -t "results/vcf/${sample}.filtered.vcf.gz"
```

Model and acceptance criteria

mpileup summarizes read evidence and likelihoods. DP records sample depth; AD records reference and alternate support. call --ploidy 1 uses a haploid model; -m selects the multiallelic caller and -v emits variant sites only.

VCFtools retained QUAL >= 30, sample FORMAT/DP >= 5 and nonmissing genotypes. minDP masks low-depth genotypes; max-missing 1 then excludes missing calls in these single-sample files. view -Oz writes BGZF VCF and index -t creates a Tabix index.

Sample	Raw	Filtered	Removed
4863	28	22	6
4866	792	710	82
9044	12	8	4

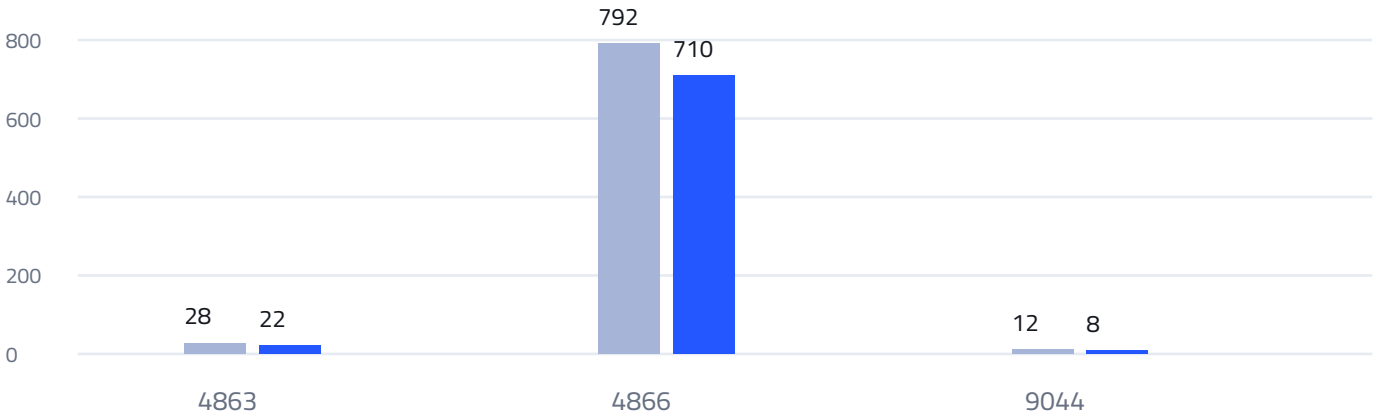
Basis: VCF headers, filtering logs and record recounts. Retained records met these criteria; filtering is not experimental confirmation.

08 / Classifying variant records

```
bcftools stats "results/vcf/${sample}.filtered.vcf.gz" \
  > "reports/${sample}.filtered.stats.txt"
bcftools query -f '%REF\t%ALT\n' \
  "results/vcf/${sample}.filtered.vcf.gz" | \
  awk 'length($2)>length($1){i++}
       length($2)<length($1){d++}
       END{print "Insertions",i+0;print "Deletions",d+0}'
```

BCFtools stats classified SNPs and indels. Comparing REF/ALT lengths separated insertions and deletions in these single-alternate-allele records. The counts describe records, not total bases changed.

Sample	Total	SNPs	Insertions	Deletions
4863	22	18	3	1
4866	710	614	57	39
9044	8	6	1	1



Raw (gray) and filtered (blue) variant records; counts from VCF files.

Output: reports/variant_summary.tsv and reports/<sample>.filtered.stats.txt. The 39 deletion records in 4866 are small VCF events and do not include the large coverage gap.

09 / Comparing exact alleles

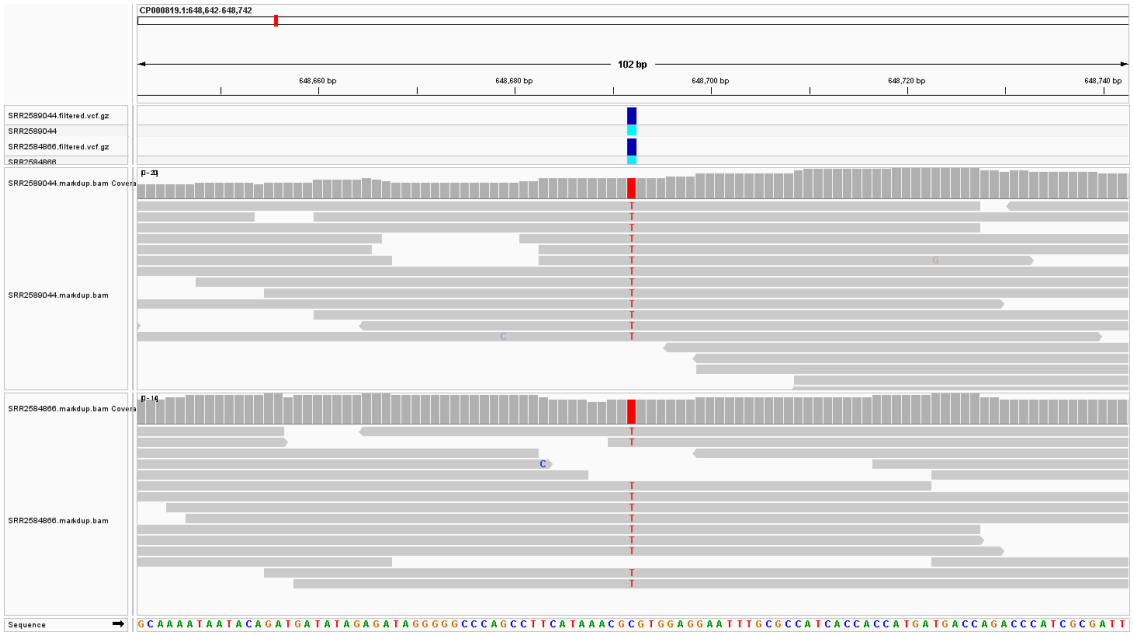
```
bcftools norm -f "$ref" -Oz \
  -o "results/vcf/${sample}.normalized.vcf.gz" \
  "results/vcf/${sample}.filtered.vcf.gz"
bcftools index -t "results/vcf/${sample}.normalized.vcf.gz"
bcftools isec -n=3 -c none \
  results/vcf/SRR2584863.normalized.vcf.gz \
  results/vcf/SRR2584866.normalized.vcf.gz \
  results/vcf/SRR2589044.normalized.vcf.gz \
  > reports/shared_all_three.tsv
```

norm standardized indel representation against the reference. isec -n=3 required presence in all inputs; -c none required identical REF/ALT alleles. Pairwise comparisons used -n=2, and single-sample-only calls used -n=1 across all three.

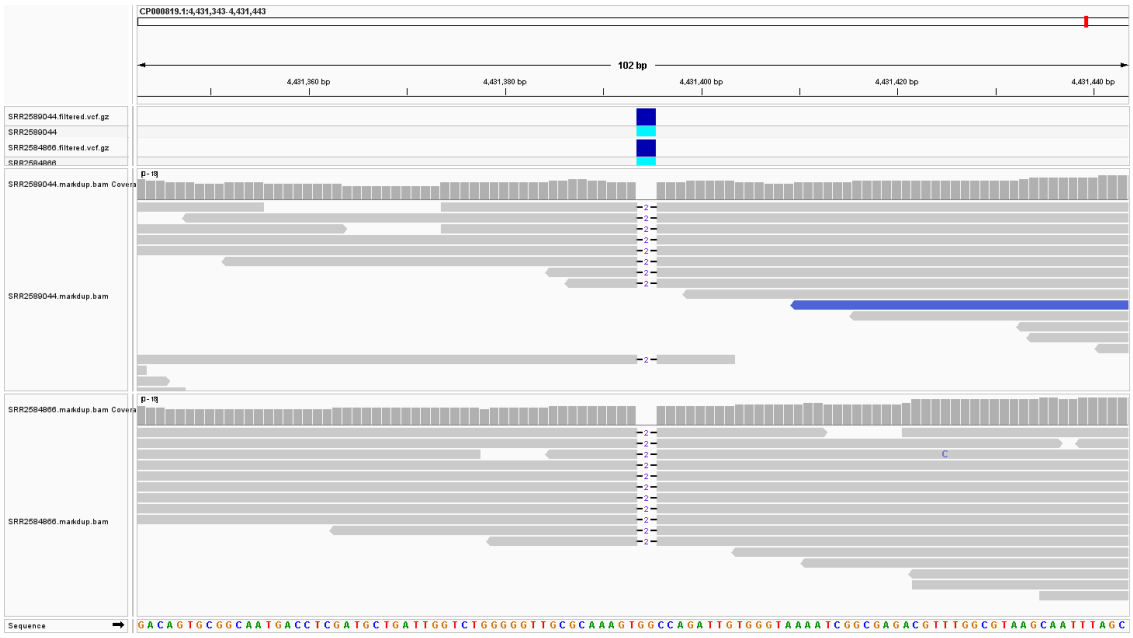
Position	REF	ALT	Gene
648692	C	T	mrdB
1331794	C	A	topA
1733343	G	A	pykF
3481820	A	G	malT
4100183	A	G	hslU
4431393	TGG	T	ytfN

Shared: 6. Pairwise overlap: 4863/4866 = 19; 4863/9044 = 6; 4866/9044 = 6. Unique filtered records: 3, 691 and 2 (696 total). Gene names were assigned in the later GFF analysis.

10 / Visual evidence for shared variants



A / Shared C-to-T SNP at 648692. Original IGV comparison.

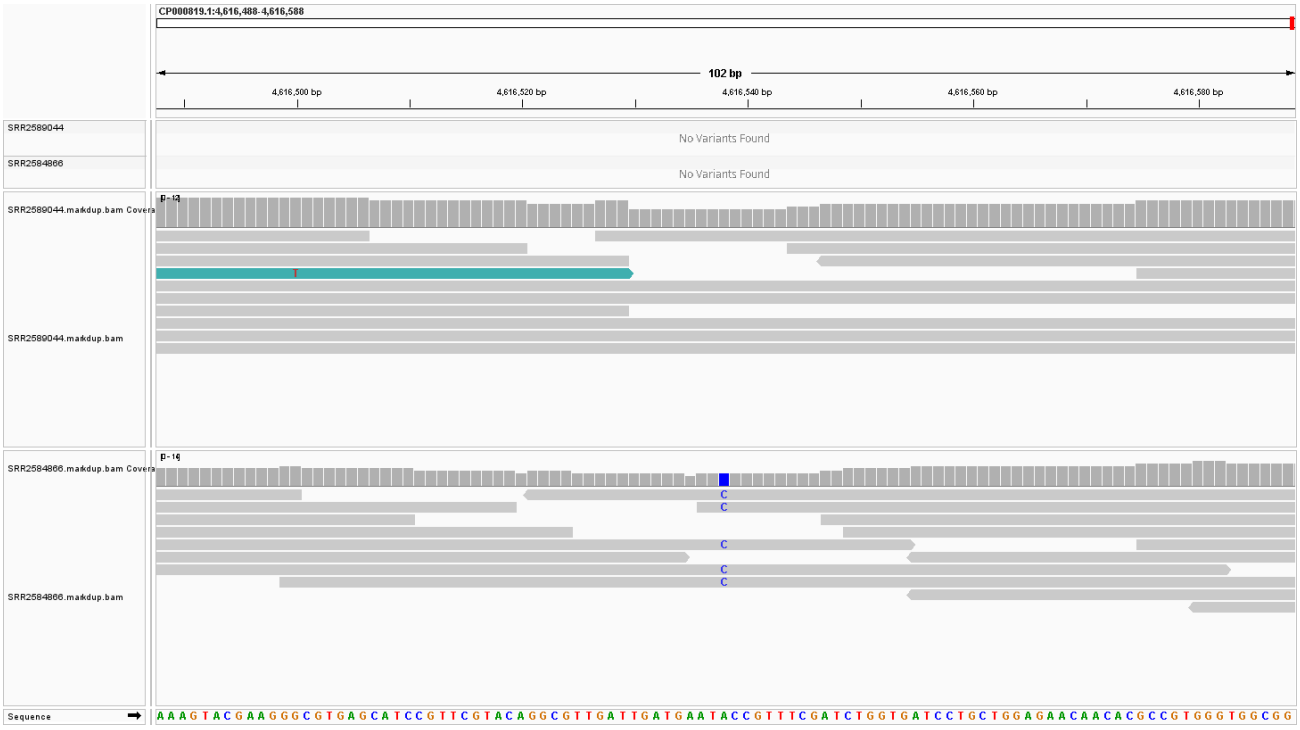


B / Shared TGG-to-T call at 4431393. The anchor T remains; GG at 4431394-4431395 is deleted.

IGV loaded the same REL606 FASTA, final indexed BAMs and filtered VCFs. Read alignments and local coverage were inspected alongside the called alleles. These figures provide selected visual evidence, not genome-wide validation.

```
reports/IGV_shared_SNP_648692.png
reports/IGV_shared_deletion_4431393.png
```

11 / Why filtered absence is conditional



Original IGV comparison at 4616538. Displayed read counts can differ from VCF depth because of counting/display settings.

Sample	Raw allele	DP	AD	Filtered
4863	A > C	9	0,9	Retained
4866	A > C	4	0,4	Excluded

The same allele was called in both raw VCFs, but 4866 failed DP >= 5. Apparent uniqueness in a filtered set does not establish biological absence in another sample.

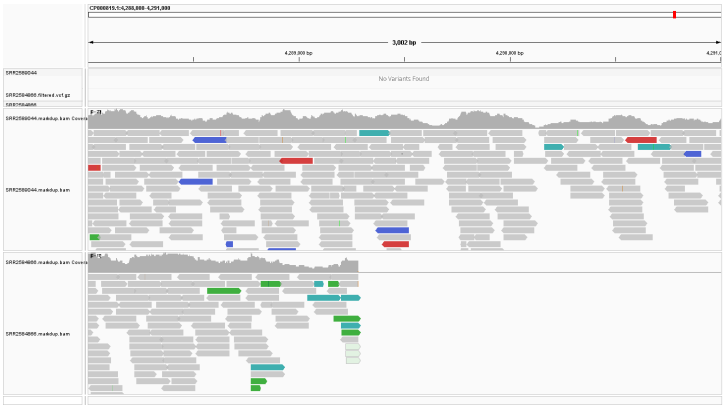
Mixed support also occurred: 4863 had AD=3,5 at 2407766 and AD=8,3 at 2618472. A haploid GT is a model-based call and does not alone demonstrate fixation in the whole population.

```
bcftools norm -f "$ref" -Oz \
  -o "results/vcf/${sample}.raw.normalized.vcf.gz" \
  "results/vcf/${sample}.raw.vcf.gz"
bcftools query \
  -f '%POS\t%REF\t%ALT[\t%GT\t%DP\t%AD]\n' \
  "results/vcf/${sample}.raw.normalized.vcf.gz"
```

12 / Investigating the coverage gap

```
samtools depth -aa results/bam/SRR2584866.markdup.bam |
awk '$3==0 {if(!run){chr=$1;start=$2} end=$2;run++;next}
run {print chr,start,end,run;run=0}
END {if(run) print chr,start,end,run}' |
sort -k4,4nr | head -n 10
```

depth -aa includes zero-depth positions; awk groups contiguous runs and sort ranks them by length. SRR2584866 had zero depth from 4,289,284 to 4,333,227 inclusive: 43,944 bases under the applied SAMtools settings.



A / Left boundary, original IGV image.



B / Right boundary, original IGV image.

The observed gap is consistent with a region requiring structural investigation. No specific deletion junction was established, so it was not counted as a confirmed deletion or added to the small-variant totals.

13 / Gene annotation of shared variants

The six shared alleles were mapped to gene and CDS features in the NCBI GFF for the same REL606 assembly. This extension did not change alignment, calling or filtering.

```
url="$base/$assembly/${assembly}_genomic.gff.gz"
curl -fL -o data/ref_genome/ecoli_rel606.gff.gz "$url"
gunzip -k data/ref_genome/ecoli_rel606.gff.gz
awk -F '\t' '
NR==FNR {pos[++n]=$2;ref[n]=$3;alt[n]=$4;next}
$1=="CP000819.1" && ($3=="gene" || $3=="CDS") {
  for(i=1;i<=n;i++) if($4<=pos[i] && $5>=pos[i])
    print pos[i],ref[i],alt[i],$3,$4,$5,$7,$9
}' OFS='\t' reports/shared_all_three.tsv \
  data/ref_genome/ecoli_rel606.gff \
  > reports/shared_variants_annotation.tsv
```

The overlap test was feature start <= variant position <= feature end. Both deleted bases at 4431394-4431395 also lie inside the ytfN CDS. Attribute extraction retained the supplied gene, locus tag, strand and product.

Gene	Locus tag	Strand	Annotated product
mrdB	ECB_00603	-	Cell wall shape protein
topA	ECB_01250	+	DNA topoisomerase I
pykF	ECB_01645	+	Pyruvate kinase
malT	ECB_03270	+	Regulator MalT
hslU	ECB_03816	-	Protease ATP-binding subunit
ytfN	ECB_04092	+	Hypothetical protein

Saved summary: reports/shared_variants_gene_summary.tsv. Product labels above are shortened for layout; the original GFF wording is retained in the TSV.

14 / Predicted coding consequences

5

MISSENSE

1

FRAMESHIFT



Gene	Position	Codon change	Predicted protein change
mrdB	648692	CGC > CAC	p.Arg69His
topA	1331794	ACG > AAG	p.Thr792Lys
pykF	1733343	GAT > AAT	p.Asp127Asn
malT	3481820	ACC > GCC	p.Thr46Ala
hslU	4100183	TCT > CCT	p.Ser350Pro
ytfN	4431393	GGC > CCA*	p.Gly1176ProfsTer5

* For ytfN, the codon pair is the first altered codon, not the complete shifted sequence. Each prediction considers the individual variant on the reference CDS; complete sample-specific protein haplotypes were not reconstructed.

These are sequence-consequence predictions. They do not demonstrate altered enzyme activity, loss of function, increased fitness or selection. The DNA/protein illustration is generated conceptual art, not a computed structure of these proteins.

```
reports/shared_variants_protein_effects.tsv
reports/<gene>_<position>_protein_effect.tsv
```

15 / Codon checks and strand direction

For a continuous phase-0 CDS, coding offset is POS minus CDS start on the positive strand, or CDS end minus POS on the negative strand. The amino-acid index is floor(offset/3)+1. Codons were retrieved with SAMtools faidx and translated in coding direction.

```
samtools faidx "$ref" CP000819.1:1733343-1733345
# pykF: GAT > AAT; p.Asp127Asn
samtools faidx "$ref" CP000819.1:1331793-1331795
# topA: ACG > AAG; p.Thr792Lys
samtools faidx "$ref" CP000819.1:3481820-3481822
# malT: ACC > GCC; p.Thr46Ala
samtools faidx -i "$ref" CP000819.1:648691-648693
# mrdB: CGC > CAC; p.Arg69His
samtools faidx -i "$ref" CP000819.1:4100181-4100183
# hslU: TCT > CCT; p.Ser350Pro
```

Negative-strand genes require complementation

faidx -i returns the reverse complement. VCF REF/ALT remain reference-forward, so the alternate base must also be complemented when interpreting a negative-strand CDS. For example, mrdB genomic C>T becomes coding G>A within CGC, producing CAC.

Gene	CDS start	CDS end	Strand
mrdB	647785	648897	-
topA	1329420	1332017	+
pykF	1732965	1734377	+
malT	3481685	3484390	+
hslU	4099899	4101230	-

Basis: recorded GFF/CDS coordinates, faidx sequence outputs and saved protein-effect TSV summaries. These were manual sequence checks rather than an automated whole-genome effect annotation.

16 / The ytfN frameshift

The VCF allele TGG>T at 4431393 retains anchor T and removes GG at 4431394-4431395. The positive-strand CDS spans 4427869-4431648. Two deleted bases shift the frame because two is not divisible by three.

```
samtools faidx "$ref" CP000819.1:4431394-4431423
# GGCCAGATTGTGGGTAAAATCGGCGAGACG
```

Reference, beginning at codon 1176

GGC CAG ATT GTG GGT

Delete the initial GG and regroup

CCA GAT TGT GGG TAA

Pro Asp Cys Gly STOP

The first altered amino acid is residue 1176. TAA is the fifth codon in the altered frame, at residue 1180. The predicted consequence is p.Gly1176ProfsTer5, a frameshift followed by an early stop.

This describes the local sequence consequence of this deletion alone. A functional knockout or a measured phenotype was not demonstrated. The GFF product is retained as hypothetical protein.

```
reports/ytfN_4431393_protein_effect.tsv
reports/shared_variants_protein_effects.tsv
```

17 / Deliverables and interpretation limits

Per-sample analysis files

```
results/bam/<sample>.markdup.bam
results/bam/<sample>.markdup.bam.bai
results/vcf/<sample>.filtered.vcf.gz
results/vcf/<sample>.filtered.vcf.gz.tbi
results/vcf/<sample>.normalized.vcf.gz
results/vcf/<sample>.normalized.vcf.gz.tbi
```

Reference, reports and visual evidence

```
data/ref_genome/ecoli_rel606.fasta
data/ref_genome/ecoli_rel606.fasta.fai
results/qc/raw/multiqc_report.html
reports/variant_summary.tsv
reports/<sample>.flagstat.txt
reports/<sample>.coverage.tsv
reports/<sample>.filtered.stats.txt
reports/shared_all_three.tsv
reports/shared_variants_gene_summary.tsv
reports/shared_variants_protein_effects.tsv
reports/IGV_*.png
```

Retain raw VCFs, intermediate BAMs, comparison tables and IGV sessions for traceability. XML sessions store local paths and need the underlying files to reopen elsewhere. Six IGV figures include the 9972 comparison and both gap boundaries.

Original file review validated archive integrity, complete SAM/BAM parsing, coordinate order and indexed retrieval, and recalculated alignment and variant summaries. MQ header typing produced a warning; it was not itself evidence of a failed call. A raw 4863 VCF index was absent from the reviewed package.

Conclusion: the required alignment, variant and visual evidence outputs were produced. Results remain conditional on subsampling, approximately 10x mean depth and the stated filters. No functional experiment or confirmed structural deletion was reported.